

Ruoshui Li

liruoshui0710@gmail.com | +1 (959) 929-1143 | github.com/Walter-Li17/academic-projects

Education

University of Connecticut Sept. 2020 – Dec. 2024
B.A. in Mathematics; Math GPA: 3.7/4.0

- Major coursework included Abstract Algebra, Advanced Abstract Algebra, Topology, Analysis, and Partial Differential Equations.

East China Normal University (ECNU) Sept. 2020 – Dec. 2024
Mathematics studies

Research Interests

Algebra, partial differential equations, real analysis, measure theory, topology.

Teaching

MTH 1400 Precalculus Spring 2026
Teaching and course-material preparation

- Designed and typeset quizzes, review exam materials, and final exam materials for a precalculus course.
- Prepared assessment documents, answer materials, and organized course PDFs using LaTeX and structured file workflows.
- Built a consistent archive of quiz and exam materials for student review and future teaching use.

Selected Projects

Product Measure for Fubini and Tonelli Theorems 2026
Measure theory project

- Wrote an annotated exposition of product sigma-algebras, product measure, Tonelli's theorem, and Fubini's theorem.
- Included examples and proof explanations aimed at readers with a first background in measure theory.

Project 8: Multivariable Taylor Theorem 2026
Multivariable analysis project

- Developed a written project and presentation explaining multivariable Taylor's theorem, directional derivatives, the proof by reduction to one variable, and applications to error estimates and critical points.
- Prepared classroom-style slides with theorem statements, examples, and space for handwritten proof discussion.

Calculating Normal Probabilities without Tables Fall 2025
Graduate project for MATH 5021, Saint Louis University

- Presented an exposition of Bagby's 1995 approximation method for standard normal probabilities.
- Explained the transition from a one-dimensional normal integral to a double integral and then to polar coordinates.
- Prepared both slide and poster materials for mathematical communication.

Direct Reading Program, University of Connecticut Dec. 2024
Independent reading project

- Studied modules and examples using *Abstract Algebra* by Dummit and Foote, Chapter 10.
- Delivered a 15-minute classroom presentation based on independent reading and mentor feedback.

Skills

Languages Mandarin Chinese (native), English (proficient).
Technical LaTeX, basic Python programming, Microsoft Word, Excel, PowerPoint, and mathematical document preparation.
Mathematics Algebra, topology, analysis, measure theory, partial differential equations, and probability exposition.

Selected Textbooks Studied

Allen Hatcher, *Algebraic Topology*, Chapter 1; David S. Dummit and Richard M. Foote, *Abstract Algebra*; Sheldon Axler, *Linear Algebra Done Right*; John Gamelin and Robert Greene, *Introduction to Topology*; William E. Boyce and Richard C. DiPrima, *Elementary Differential Equations and Boundary Value Problems*; Stanley J. Farlow, *Partial Differential Equations for Scientists and Engineers*; J. W. Brown and R. V. Churchill, *Complex Variables and Applications*; Jay Cummings, *Real Analysis*; Thomas W. Judson, *Abstract Algebra: Theory and Applications*.